$$\hat{U}(t) = \mathrm{e}^{\mathrm{i}\phi(t)} \exp\left[-\mathrm{i}\int_0^t \mathrm{d}t'\, \hat{H}_{\mathrm{int}}(t')\right] \equiv \mathrm{e}^{\mathrm{i}\phi(t)}\hat{V}(t). \tag{5.98}$$